

Indian Institute of Information Technology, Allahabad



Mid-Semester Project
Evaluation

KENT-AI

An AI-Powered Conversational Clinical Assistant
for Homeopathic Case-Taking and Dense
Semantic Repertorization

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Introduction

Homeopathy is an individualized medical science that treats patients holistically based on the Law of Similars. To identify the curative remedy (Simillimum), physicians rely on repertorization—the systematic process of mapping patient suffering to standardized repertory rubrics.

Homeopathic Principles & Repertorization

- **Law of Similars:** Similia Similibus Curentur ("Like cures like") — micro-diluted remedies matching the patient's individual totality.
- **What is Repertorization?:** The systematic translation of free-text patient complaints into standardized clinical taxonomy entries called **Rubrics**.
- **The Simillimum:** The single indicated remedy that covers the symptom totality across physical, modal, and mental spheres.

Kent repertory:

- **Classical Foundation:** Compiled by **Dr. J. T. Kent (1897)**; indexes **74,513** rubrics across **37** anatomical sections.
- **Database Scale:** **679** remedies and **507,179** rubric-remedy associations.

Kent's 3-Tier Typographic Grading System:

-  **Grade 3 (Bold Capitals, Weight = 3):** Highest clinical prominence; verified across multiple provers and clinical cures (313,519 links).
-  **Grade 2 (Italics, Weight = 2):** Moderately verified; confirmed by independent provers (113,336 links).
-  **Grade 1 (Roman/Plain, Weight = 1):** Solitary proving symptoms or occasional clinical observations (31,499 links).

Motivation

The Clinical Consultation Bottleneck

- **Time-Intensive Case-Taking:** 30–45 minutes per consultation limits capacity.
- **Cognitive Fatigue & Bias:** 74,513 rubrics are impossible to cross-reference manually.
- **Inter-Practitioner Inconsistency:** Different doctors may extract different rubrics from the same case.

The Semantic & Lexical Divide:

Patient Expression	Kent's 19th-Century Rubric	Keyword Overlap
"Crushing vice-like headache"	HEAD > PAIN > pressing > as if in a vise	0% Overlap
"Worse when bending over"	HEAD > PAIN > stooping, from	0% Overlap
"Throbbing above right eyebrow"	HEAD > PAIN > forehead > right > over eyes	0% Overlap
"Fear of being alone at night"	MIND > FEAR > alone, of being > night	0% Overlap

Why Dense Vector Search is Mandatory: Colloquial modern English and Victorian inverted syntax have zero lexical overlap; exact SQL/FTS queries fail completely.

Research Problems & Core Challenges

Pillar 1 – 7D Clinical NER:

Extracts Location, Sensation, Modalities, Concomitants, Temporal, and Mental/Emotional features from complex clinical **dialogue**.

Pillar 2 – Taxonomic Retrieval:

Maps extracted symptoms to the optimal rubric across **74,513** candidates, preserving hierarchical context with **<15 ms** retrieval latency.

Pillar 3 – Data Scarcity:

No public token-annotated dataset exists for clinical symptom extraction; manually annotating **20,000+** cases is financially and logistically impractical.

Literature Review

AI in Homeopathy (2024–2026):

HOHM Foundation (Healthcare, 2026) →

Four LLM chatbots matched the practitioner's remedy in only 36.5% of cases, and repeated queries gave different answers. <https://www.mdpi.com/2227-9032/14/7/909>.

HomeoCure (Springer, LLM-RAG) →

It retrieves symptom-remedy pairs from a digitized homeopathic text and asks follow-up questions for incomplete symptoms. https://link.springer.com/chapter/10.1007/978-3-032-23241-0_6

Literature Review (Continued)

Clinical NLP & Dense Retrieval (2019-2026)

Bio_ClinicalBERT (Alsentzer et al., 2019) →

Domain-specific pretraining on clinical text improved performance on 3 of 5 clinical NLP tasks.

It also set a new state of the art on MedNLI. <https://doi.org/10.18653/v1/W19-1909>.

MediTOD (Saley et al., EMNLP 2024) →

A history-taking dialogue dataset annotated with medical slots. Its attributes, such as onset and severity, support structured symptom intake. <https://aclanthology.org/2024.emnlp-main.936/>

Research Gap Analysis

Feature	Traditional Tools (RadarOpus)	General AI (ChatGPT)	Research AI (HomeoCure)	Kent-AI (Our Work)
Search Method	Exact keywords	Free-text generation	Text similarity search	Conversational FSM + Vector Search
Database Grounded	✓ Yes (Static)	✗ No (Hallucinates)	⚠ Partial text	✓ 100% Grounded (74,513 Rubrics)
Symptom Extraction	✗ Manual by doctor	✗ Unstructured prose	⚠ Simple text	✓ Bio_ClinicalBERT (7 Dimensions)
Automated Intake	✗ None	✗ Freeform chat	⚠ Basic questions	✓ Structured 10-State Intake Bot
Remedy Ranking	Simple addition	✗ Inconsistent output	⚠ Basic scoring	✓ Kentian Grade-Weighted + IRF Penalty
Doctor Dashboard	Complex legacy UI	✗ None	✗ None	✓ Modern Google Stitch Dashboard
Privacy & Cost	Local; ₹23,000-₹1.5 Lakhs	Cloud API (Privacy Risk)	Lightweight model	Local-first: runs on clinician's PC (Ollama, SQLite, ChromaDB); no license fee

Problem Statement

1. The Clinical Problem

- Homeopathic consultations take **30 to 45 minutes**, causing physician fatigue and capping clinic capacity.
- Patients speak **modern** everyday language, while classical repertories use **19th-century medical terms**, leading to **0%** keyword overlap.
- Doctors cannot remember **74,513** rubrics, leading to cognitive bias toward **~30** common remedies.

2. The Computational / AI Problem

- **Zero labeled training data** exists for homeopathic clinical entity recognition.
- Unconstrained generative AI (like ChatGPT) **hallucinates fake remedies** and lacks database provenance.
- Cloud-based AI poses serious **patient data privacy (PHI)** risks in real clinic workflows.

3. The Clear Goal Statement

To develop an **offline**, privacy-preserving clinical assistant that automatically extracts Kent's **7 symptom dimensions** through structured conversational intake, and delivers **100% database-grounded** remedy repertorization in under 10 minutes

Existing Knowledge Base

Our project leverages and builds upon the digitized Kent's Repertory open-source foundation developed by su4532 (github.com/su4532/kent-repertory-explorer).

- This provides an initial structured **SQLite** schema with **74,513** rubrics, **679** remedies, and **507,179** mappings.
- Having this raw digital foundation in place allows our team to focus directly on the core machine learning challenges: **1)** dense vector representation, **2)** ClinicalBERT symptom NER, **3)** synthetic case generation, and **4)** conversational clinical reasoning.

Kent's Repertory: **37** chapters organized head-to-foot.

- **Selected Chapter 1 — MIND:** 4,933 rubrics as our initial benchmark.
- **Why MIND?**
 - Mental symptoms are given **highest** importance in classical case analysis.
 - Uses **abstract & figurative language**, making it the hardest AI reasoning challenge.
 - **4,933** MIND rubrics give us a solid, focused dataset to generate **~22,200** synthetic cases and train **Bio_ClinicalBERT** before expanding to physical chapters.
- **Next:** Expand from **MIND** → **physical chapters** after validation.

Kent's 7 Clinical Symptom Dimensions

Dimension	What it captures	Example
Location	Body part/region	"Right side of head"
Sensation	Type of feeling	"Pressing, throbbing, burning"
Modality	Aggravation/amelioration	"Worse in morning, better by pressure"
Laterality	Left/right/bilateral	"Left-sided"
Concomitant	Accompanying symptoms	"With nausea and anxiety"
Mental/Emotional	Psychological state	"Irritable, restless at night"
Temporality	Time patterns	"Worse at midnight, periodic"

Why This Matters for Kent-AI:

- **The "Complete Symptom" Principle:** In classical homeopathy, a raw complaint like "**I have a headache**" cannot be prescribed upon. It must be qualified across all **7** dimensions (location, sensation, triggers, and emotional state) to find the true Simillimum.
- **The Target Schema for Our AI:** Our **10-State Conversational FSM** and Bio_ClinicalBERT are specifically engineered to extract, track, and validate these exact 7 dimensions from everyday patient dialogue.

(Greeting→Chief Complaint→Location→Sensation→Worse From→Better From→Associated Symptoms→Mind & Mood→Review→Done)

Formal Mathematical Problem Formulation

Intake & Extraction

1. Intake StateTransition

$$S_{t+1} = \delta(S_t, u_t, \Phi_t), S_t \in S_{\text{intake}}, \Phi_t \in [0,1]^7$$

Moves to the next question based on the patient's reply and which of the 7 dimensions are already filled.

2. Token-Level Entity Recognition

$$y^i = \operatorname{argmax}_{c \in Y} P(y_i = c \mid x; \theta^{\text{BERT}}), \\ |Y| = 15 \text{ * BIO Tags}$$

Predicts whether each word belongs to Location, Sensation, Modality, Time, or Mind.

In **NLP**, **BIO** stands for **B**eginning, **I**nside, **O**utside. It is the standard way AI models tag exact word boundaries for entities in a sentence.

Kent-AI extracts **7** symptom dimensions, each using **B- & I- tags, plus O (Outside) → 15 BIO tag**

Vector Search & Totality Scoring

3. Dense Semantic Retrieval:

$$\text{Sim}(q_k, e_r) = [(q_k \cdot e_r)] / (\|q\|_2 \|e\|_2), r \in R_{74,513}$$

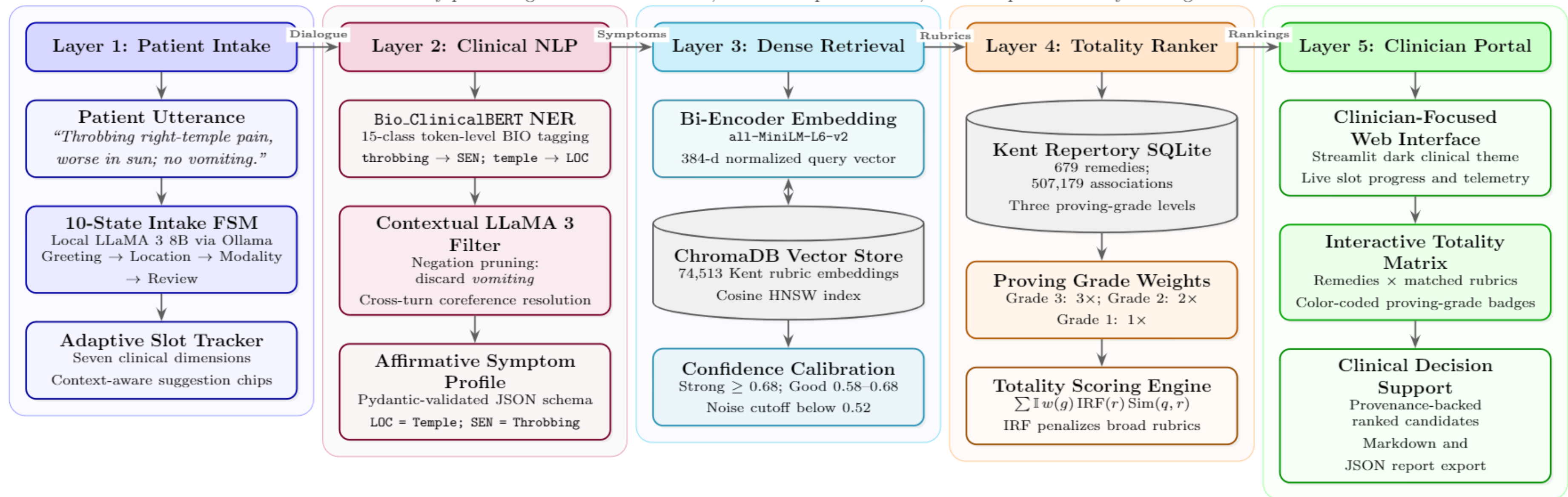
Calculates the cosine similarity between the patient's symptom query and all 74,513 pre-computed rubric vectors.

4. Kentian Totality Scoring

$$\text{Score}(m \mid R^*) = \sum_{r \in R^*} \mathbb{I}(m \in M_r) \cdot w(\text{gr}, m) \cdot \text{IRF}(r) \cdot \text{Sim}(q_r, r)$$

Combines rubric coverage (**I**), proving grade weight (**w=3,2,1**), search similarity (**Sim**), and an Inverse Remedy Frequency (**IRF**) penalty to stop common remedies from crowding out specific ones

Proposed End-to-End System Architecture



- **High-Speed Execution** — **<15 ms** semantic search across **74,513** rubrics; reduces review time from **45 min** to **<10 min**.
- **Grounded & Traceable** — Every recommendation is **mathematically** tied to Kent's classical grades (3, 2, 1).

Conversational Intake Agent & 10-State FSM



Three Smart Capabilities:

- **Auto-Fills Multiple Details:** If a patient says "**Burning pain in my right temple worse from light**", the system fills Location, Sensation, and Triggers at once, skipping redundant questions.
- **Empathetic Bedside Prompting:** Local **LLaMA 3** is instructed to sound supportive and clear without drifting off-topic.
- **Dynamic Quick-Reply Chips:** Generates **clickable** suggestion chips (e.g., **[Worse in morning]**, **[Throbbing pain]**) so mobile and elderly patients do not have to type long paragraphs.

Symptom Extraction & LLaMA 3 Resolution Layer

1

Token-Level Bio_ClinicalBERT Tagging

Uses a 15-class BIO tagging scheme to identify exact word spans for Location, Sensation, Aggravation, Relief, Concomitants, Time and Mind.

Patient Input (Raw Text)

“Severe throbbing pain in right temple worse in morning, but no vomiting or dizziness.”



Tokens with BIO Tags (15 Classes)

Token	BIO Tag
Severe	B-SEN
throbbing	I-SEN
pain	I-SEN
in	O
right	B-LOC
temple	I-LOC
worse	B-MOD_AGG
in	O
morning	B-TEMP
but	O
no	O
vomiting	O
or	O
dizziness	O

2

LLaMA 3 Contextual Resolution

Filters negations, resolves pronouns, and produces a clean structured profile.



1. Negation Filtering

Drops negated statements.

"no vomiting or dizziness"

→ marked negative and ignored.

"no vomiting"

negated → ignored

"dizziness"

kept (if not negated)



2. Pronoun Resolution

Connects vague words to correct entity.

"it radiates downward"

→ linked to "temple".

"it"

refers to

"temple"



3. Clean Structured Profile

Emits structured JSON ready for vector search.

Only valid, resolved symptoms are kept.

Resolved Symptom Profile (JSON)

```
{
  "symptoms": [
    {
      "text": "severe throbbing pain",
      "type": "SEN",
      "location": "right temple",
      "modality": {
        "worse": "morning",
        "better": null
      }
    }
  ],
  "negated_symptoms": [
    {
      "text": "vomiting",
      "type": "CONC",
      "reason": "negated"
    },
    {
      "text": "dizziness",
      "type": "CONC",
      "reason": "negated"
    }
  ]
}
```

Synthetic Case Generation Pipeline



Dense Semantic Retrieval & Remedy Ranking

ChromaDB Dense Vector Search

- **Corpus Size:** All **74,513 rubrics** pre-embedded using all-[MiniLM-L6-v2](#) (d=384).
- **Search Speed:** Sub-15 ms cosine search across the entire index on standard CPU.
- **Calibrated Match Tiers:**
 -  **Strong Match (≥0.68):** Clear clinical synonym.
 -  **Good Match (0.58–0.68):** High clinical relevance.
 -  **Fair Match (0.52–0.58):** Exploratory match for doctor review.
 -  **Noise Filter (<0.52):** Automatically discarded to prevent wrong matches.

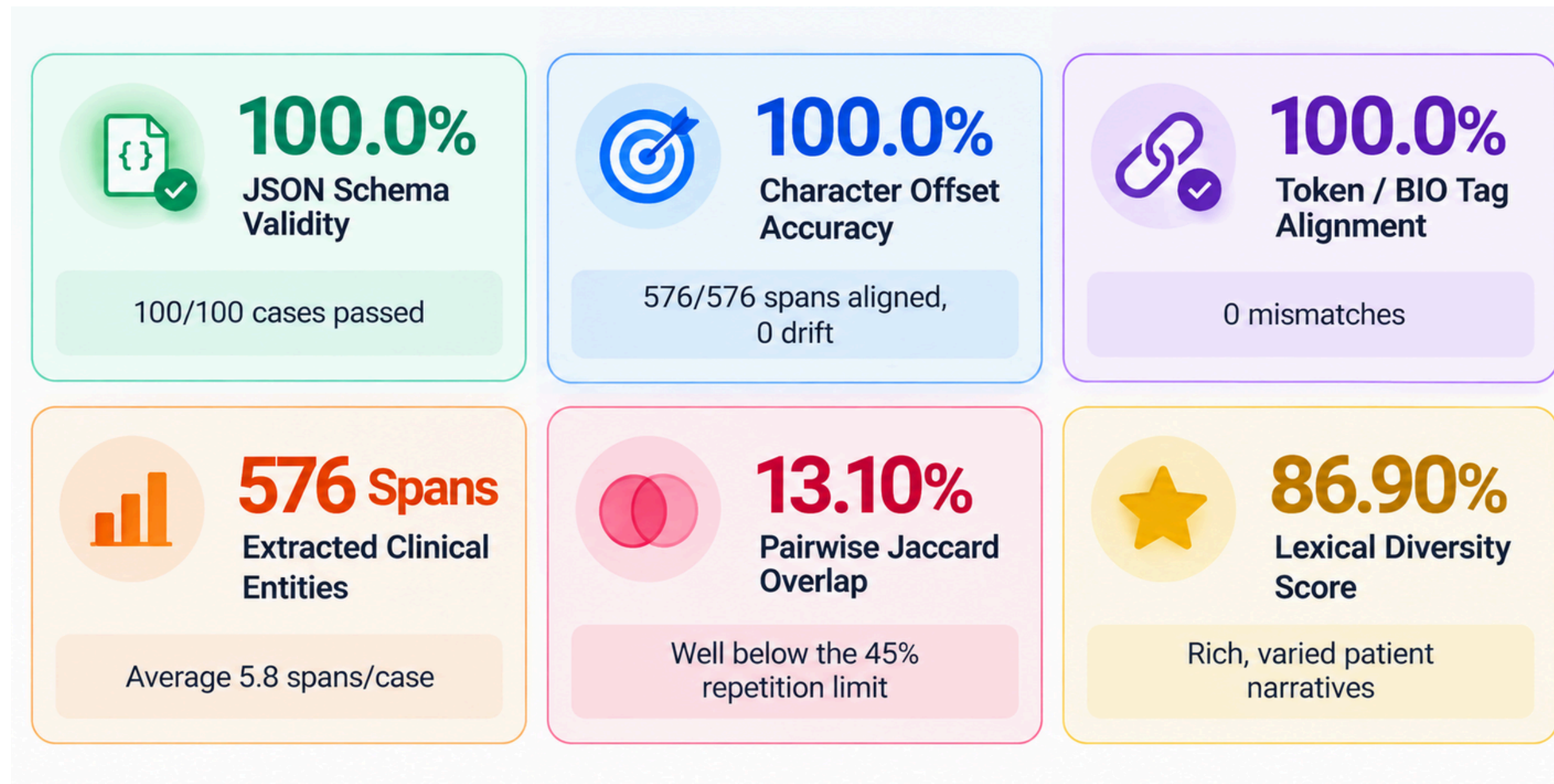
Classical Kentian Remedy Ranker

- **Scoring Logic:**
 - **Score** = $\Sigma(\text{Coverage} \times \text{Grade Weight} \times \text{IRF Penalty} \times \text{Search Similarity})$
- **Proving Weights:** Grade **3 Bold (3×)**, Grade **2 Italics (2×)**, Grade **1 Plain (1×)**.
- **IRF Specificity Penalty:** Lowers the score of hyper-common remedies (**Sulphur**), helping specific curative remedies rank on top.

Pilot Benchmark Validation Results

(100 MIND Cases)

Verified on **100** Clinical Cases Across **25** MIND Rubrics

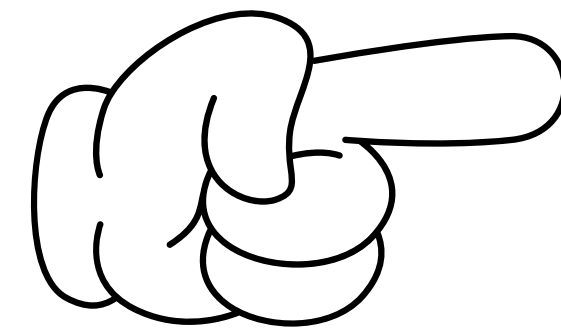


7-Dimension Breakdown:

- **MENT:** 286 (49.7%)
- **SEN:** 77 (13.4%)
- **LOC:** 76 (13.2%)
- **MOD_AGG:** 57 (9.9%)
- **TEMP:** 39 (6.8%)
- **CONC:** 34 (5.9%)
- **MOD_AMEL:** 7 (1.2%)

Clinical Dashboard

(Initial Design)





Kent-AI Clinical Assistant

Classical Homeopathic Repertorization & Decision Support

● ⓘ Session: MIND (4.9k) ⓘ

Live Patient Intake

Repertorization Engine

Rubric Explorer

Materia Medica Index



Live Symptom Dimensions

Step 1 of 9: Greeting

Totality: 0/7

Location

Not asked yet

Sensation

Not asked yet

Worse from
(aggravation)

Not asked yet

Better from
(amelioration)

Not asked yet

Concomitant

Not asked yet

Time pattern

Not asked yet

Mental/Emotional

Not asked yet



Intake Controls

Active Case Summary:

No symptoms captured yet. Converse in the dialogue console to extract dimensions.

Generate Repertorization (0/7)

Requires primary complaint + at least one modality (0/7 collected).

Reset Consultation

Intake Guidance (?) ▾



Clinical Dialogue Console

● LLaMA 3:8B Online



Hello! I'm Kent-AI, your clinical intake companion. We will examine your main complaint, including location, sensations, triggers, relief factors, and emotional state to guide classical repertorization. What primary symptoms have been troubling you most?



Suggested Responses:

Click to reply instantly

Severe anxiety & fearful apprehension

Emotional pain & sadness, worse alone

Restless pacing & irritability

Absent-mindedness & memory lapses

Tell Kent-AI how you are feeling (e.g. 'I feel anxious and restless in the evening, worse when alone')...



✓ PRIMARY SIMILLIMUM INDICATION

Belladonna (Bell.)

Covering 6 of 14 symptom rubrics (43% coverage) with a weighted totality score of 3.81.

2. Remedy Totality Ranking

Remedy Totality Ranking (Grade-Weighted Score)



3. Top Remedy Breakdown

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Rank	Remedy	Full Name	Score	Rubrics
1	Bell.	Belladonna	3.81	6/14
2	Glono.	Glonoin	4.548	5/14
3	Ars.	Arsenicum album	1.653	5/14
4	Calc.	Calcarea carbonica	4.032	4/14
5	Phos.	Phosphorus	3.015	4/14
6	Op.	Opium	2.388	4/14
7	Bry.	Bryonia alba	2.304	4/14
8	Nux-v.	Nux vomica	2.247	4/14

Post-Mid-Semester Roadmap

Four Immediate Milestones →

Full-Scale GPU Generation: Run `scripts/run_gpu_generation.sh` on college GPU servers to generate ~**22,200** synthetic clinical cases across all 4,933 MIND rubrics.

Fine-Tuning Bio_ClinicalBERT: Train the token-level sequence classifier on **Colab GPU** (Target Token **F1** $\geq 88.0\%$).

System-Wide Evaluation: Benchmark held-out test cases targeting Top-20 Rubric **Recall** $\geq 90.0\%$ and **MRR** ≥ 0.65 .

Clinical Expert Validation (Homeopathy Practitioners): Partner with **certified homeopathic doctors** to test the tool in practice—verifying remedy choices, rubric matches, and consultation time reduction (**from 40 min to <10 min**).

Conclusion

Kent-AI gives homeopathic doctors an automated intake assistant that reduces consultation time from 40 to under 10 minutes, using verified repertory data to ensure accurate remedy recommendations.

Github → <https://github.com/Krishna200608/kent-ai>

THANK YOU